



Database Management Systems

Database Modification

Anchor: Dr. Nagasundari S, Dept of CSE

Database Management Systems

Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

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Dr. Nagasundari S

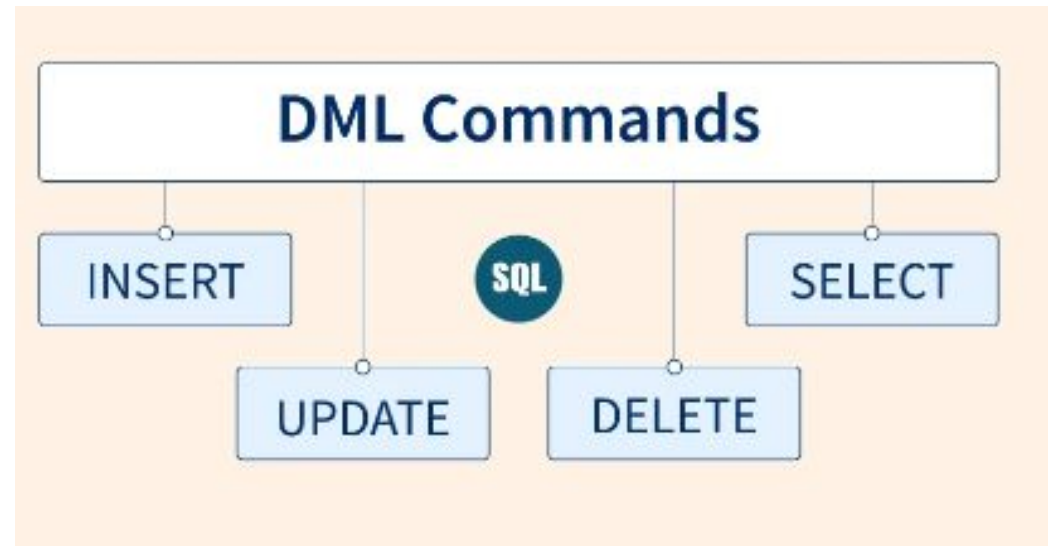
Department of Computer Science and Engineering



DML commands in Structured Query Language change the data present in the SQL database. We can easily access, store, modify, update and delete the existing records from the database using DML commands.

DML commands:

- SELECT Command
- INSERT Command
- UPDATE Command
- DELETE Command



- In its simplest form, INSERT is used to add a single tuple (row) to a relation (table).
- We must specify the relation name and a list of values for the tuple.
- There are two forms of the INSERT statement.

Form 1:

The values should be listed in the same order in which the corresponding attributes were specified in the CREATE TABLE command.

Syntax:

INSERT INTO table_name VALUES (value1,value2,.....);

Example:

```
INSERT INTO EMPLOYEE VALUES ( 'Richard', 'K', 'Marini', '653298653', '1962-12-30', '98 Oak Forest, Katy, TX', 'M', 37000, '653298653', 4 );
```

INSERT COMMAND



Before Insert:

[illegible]

Example: INSERT INTO EMPLOYEE VALUES ('Richard', 'K', 'Marini', '653298653', '1962-12-30', '98 Oak Forest, Katy, TX', 'M', 37000, '653298653', 4);

Output:

[illegible]

- Form 2 allows the user to specify explicit attribute names that correspond to the values provided in the INSERT command.
- **This is useful if a relation has many attributes but only a few of those attributes are assigned values in the new tuple.**
- However, the values must include all attributes with NOT NULL specification and no default value.
- Attributes with NULL allowed or DEFAULT values are the ones that can be left out.

Syntax:

INSERT INTO table_name(attribute1,attribute2,...) VALUES (value1,value2,.....);

Example: Enter a tuple for a new EMPLOYEE for whom we know only the Fname, Lname, Dno, and Ssn attributes

Query: INSERT INTO EMPLOYEE (Fname, Lname, Dno, Ssn) VALUES ('Richard', 'Marini', 4, '653298653');

Another variation of the INSERT command:

A variation of the INSERT command inserts multiple tuples into a relation in conjunction with creating the relation and loading it with the result of a query.

- create a temporary table that has the employee last name, project name, and hours per week for each employee working on a project

Command :

```
CREATE TABLE WORKS_ON_INFO ( Emp_name VARCHAR(15), Proj_name VARCHAR(15),  
Hours_per_week DECIMAL(3,1) );
```

Example:

```
INSERT INTO WORKS_ON_INFO ( Emp_name, Proj_name, Hours_per_week )  
SELECT E.Lname, P.Pname, W.Hours  
FROM PROJECT P, WORKS_ON W, EMPLOYEE E  
WHERE P.Pnumber = W.Pno AND W.Essn = E.Ssn;
```

Another variation: We can use the command given below to backup the data present in a table. The condition is that the schema of both the backup table and the original should must be the same.

Example:

```
INSERT INTO WORKS_ON_INFO_Backup (select * from WORKS_ON_INFO ) ;
```

Note: WORKS_ON_INFO_Backup table need to be created before using the above command

- The **SELECT** clause lists the attributes desired in the result of a query.
- It corresponds to the projection operation of relational algebra.
- **FROM** clause is used to specify the table name or the relation name.
- **WHERE** clause can be used to filter the rows

Example: Find the names of all instructors:

```
select fname,Lname
```

```
from EMPLOYEE;
```

Output:

Fname	Lname
John	Smith
John	Smith
Franklin	Wong
Joyce	English
Ramesh	Narayan
James	Borg

Note: Select will be discussed
in detail in the next lecture

- The UPDATE command is used to modify attribute values of one or more selected tuples.
- WHERE clause in the UPDATE command selects the tuples to be modified.
- However, updating a primary key value may propagate to the foreign key values of tuples in other relations if such a referential triggered action is specified in the referential integrity Additional constraints of the DDL.
- An additional SET clause in the UPDATE command specifies the attributes to be modified and their new values.
- Several tuples can be modified with a single UPDATE command.

Syntax:

UPDATE table_name SET column_name = value WHERE condition; // changes done on selected rows

UPDATE table_name SET column_name = value; // all rows are updated

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UPDATE COMMAND

Example1: To change the location and controlling department number of project number 10 to 'Bellaire' and 5, respectively.

Query: UPDATE PROJECT SET Plocation = 'Bellaire', Dnum = 5 WHERE Pnumber = 10;

Before:

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4
NULL	NULL	NULL	NULL

After:

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Bellaire	5
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4
NULL	NULL	NULL	NULL

✓	8	16:35:34	UPDATE PROJECT SET Plocation = 'Bellaire', Dnum = 5 WHERE Pnumber = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
---	---	----------	--	--

Example 1: Give all employees in the 'Research' department a 10% raise in salary.

Query: UPDATE EMPLOYEE SET Salary = Salary * 1.1 WHERE Dno = 5;

Before :

fname	salary	dno
John	33000.00	5
Franklin	44000.00	5
Joyce	27500.00	5
Richard	37000.00	4
Ramesh	41800.00	5
James	55000.00	1
Jennifer	43000.00	4
Ahmed	25000.00	4
Alicia	25000.00	4

After :

fname	salary	dno
John	36300.00	5
Franklin	48400.00	5
Joyce	30250.00	5
Richard	37000.00	4
Ramesh	45980.00	5
James	55000.00	1
Jennifer	43000.00	4
Ahmed	25000.00	4
Alicia	25000.00	4

Example 2: Give a 5 percent salary raise to employees whose salary is less than average.

Query: UPDATE employee SET Salary = Salary * 1.05 WHERE salary < (SELECT avg_sal FROM (SELECT AVG(salary) AS avg_sal FROM employee) AS temp);

Example 2: Give a 5 percent salary raise to employees whose salary is less than average.

Query: UPDATE employee SET Salary = Salary * 1.05 WHERE salary < (SELECT avg_sal FROM
(SELECT AVG(salary) AS avg_sal FROM employee) AS temp);

In PostgreSQL / Oracle / SQL Server:

UPDATE employee SET Salary = Salary * 1.05 WHERE salary < (SELECT AVG(salary) FROM employee);

This does not work in MySQL, reason is it does not allow selecting from the same table you are updating (in the same query) without wrapping it in a derived table.

- The DELETE command removes tuples from a relation.
- It includes a WHERE clause, to select the tuples to be deleted.
- Tuples are explicitly deleted from only one table at a time.
- However, the deletion may propagate to tuples in other relations if referential triggered actions are specified in the referential integrity constraints of the DDL.
- Depending on the number of tuples selected by the condition in the WHERE clause, zero, one, or several tuples can be deleted by a single DELETE command.
- A missing WHERE clause specifies that all tuples in the relation are to be deleted. However, the table remains in the database as an empty table.
- We must use the DROP TABLE command to remove the table definition.

Syntax:

DELETE FROM *table_name* WHERE *condition*

Example1: Delete the employee record whose last name is **Zelaya** .

Query: DELETE FROM EMPLOYEE WHERE Lname = 'Zelaya';

Before:

fname	lname	salary	dno
John	Smith	36300.00	5
Franklin	Wong	48400.00	5
Joyce	English	30250.00	5
Richard	Marini	37000.00	4
Ramesh	Narayan	45980.00	5
James	Borg	55000.00	1
Jennifer	Wallace	43000.00	4
Ahmed	Jabbar	25000.00	4
Alicia	Zelaya	25000.00	4

After:

fname	lname	salary	dno
John	Smith	36300.00	5
Franklin	Wong	48400.00	5
Joyce	English	30250.00	5
Richard	Marini	37000.00	4
Ramesh	Narayan	45980.00	5
James	Borg	55000.00	1
Jennifer	Wallace	43000.00	4
Ahmed	Jabbar	25000.00	4

Example2: Delete all employee records [It is similar to truncating employee table]

Query: DELETE FROM EMPLOYEE;

1. Which SQL clause is used to specify the attributes that should be displayed in the output?

- A) FROM
- B) WHERE
- C) SELECT
- D) ORDER BY

2. Which DML operation is used to remove data from a database?

- A) SELECT
- B) INSERT
- C) DELETE
- D) UPDATE

3. In the SQL DML example, "SELECT name FROM instructor WHERE dept_name = 'History'", what does 'name' refer to?

- A) A column in the 'instructor' table
- B) A table in the database
- C) A value in the 'dept_name' column
- D) A query condition

4. What is the purpose of the Data Manipulation Language (DML)?

- A) Defining the structure of a database
- B) Querying the database
- C) Managing database access
- D) Storing the data in the database

Answers:

1. C
2. C
3. A
4. B



THANK YOU

Anchor: Dr. Nagasundari S, Dept of CSE

nagasundaris@pes.edu